

DATA423 Assignment 3 Example Report

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Example Report

This example report is a suggestion of layout and style. The content is fictitious.

Be sure to include images from your Shiny App as evidence for your observations, these must have figure numbers and a detailed description.

Suggested Headings and Sub-headings maybe found in the assignment specification.

Data Description

The data consists of 900 observations of 32 variables. The target variable is “CLASS”. There are missing values in a portion of the numeric variables. The missing values visually appear random. There are no excessively missing variables or observations. The numeric predictor data has some uni-variable outliers but these disappear when the IQR multiplier reaches 1.9 so these are not of great significance. There are several nominal variables. One which, “HTYPE” has a cardinality of 62, the other, “District”, has low cardinality. The numeric data has a variety of scales. The means range from -4 to 122 across the variables. There is a similar variety in the variances (16 to 366). The predictor correlation shows that a block of numeric variables are highly correlated. These all have similar variable names i.e. “Survey_*”. Several date formatted character variables exist. The format is unambiguously mm-dd-YYYY. These have been converted to date variables. The pairs plots show some non-linear relationships between the predictors and the target.

Strategies

Missing data

There are no excessively missing variables or observations to discard. Because we have a raw obs/variables ratio of only 17 we do not in a strong position to choose partial deletion. There is NO evidence of missing values being “Missing Completely at Random” so partial deletion might generate bias. The methods that implicitly handle missing values can be tried on the raw data. For the other methods we shall employ **knn** (neighbours=5) imputation as a standard approach. Once selected, the best model was improved slightly by using bag imputation. Since there are no missing target values we do not need to discard these observations.

Outliers

The uni-variable outliers are of little concern. The observations are all to be retained. Some robust methods were tried just in case they offered some advantage – which they did not. The critical issue was the significance of model-residual outliers.

Feature engineering

Centring and scaling is suggested due to the variety in means and variances of the numeric data. It is not clear whether this will be better before or after knn imputation. Once selected, the best model was improved slightly by preceding knn imputation with centring and scaling.

All date variables were converted to decimal, day-of-week, month, and year. From variable importance it became apparent that month was a significant predictor. This suggests a seasonality component to the model's behaviour.

High cardinality variables were turned into numeric variables by using target-encoding. Low cardinality nominal variables (incl. day-of-week, month) were treated with dummy encoding.

Missing values (and future missing values) can be dealt with using imputation. Experimentation with the slow "bag imputation" was reserved for the best performing models.

The correlation between numeric variables can be eliminated by employing processing with PCA, PLA, ICA techniques.

Tuning & Assessing

A static test set of 30% was set aside to assess the best model using stratified sampling based upon the target. The hyper-parameters were tuned by using 25 bootstrap re-samples of the training data.

Methods

The large number of variables present after dummy encoding was reduced by emphasizing methods with implicit feature selection. In choosing methods to try, the strategy was based on domain precedence which historically favoured non-linear models. In case this data set was not typical of past practice, a number of wild-card methods were also tried. Generally, if a method performed badly, methods similar to it were disfavoured but not disqualified i.e. they were less likely to get tried.

In order to fairly assess a method, it was necessary to give careful consideration to the pre-processing that was likely to suit the method best. Particularly inappropriate pre-processing generally caused the training to fail.

Trials

See Table 1 for the methods tried:

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## Warning: package 'kableExtra' was built under R version 4.4.3
```

Table 1: Methods tried.

Methods	Characteristics	Notes	Reason_chosen
ANFIS - Adaptive- Network-Based Fuzzy Inference System	Rule-Based Model	Failed when missing still present. Failed when nominal still present. 2 hyper-parameters	Random example of neural network based method
bagEarth - Bagged MARS	Multivariate Adaptive Regression Splines Ensemble Model. Implicit Feature Selection. Bagging. Accepts Case Weights	2 hyper-parameters, Slow to train	Random example that copes with non-linearity and does feature selection
elm - Extreme Learning Machine	Neural Network	2 hyper-parameters. Slow to Train	Totally random
treebag - Bagged CART	Tree-Based Model. Ensemble Model. Bagging. Accepts Case Weights	Crashed 'ipred' package will not install	Random example of a tree based method
...	...	...	...

Models

The following models were successfully trained. A visual summary of the models can be seen in Figure 1. Models that performed worse than the null model have been omitted.

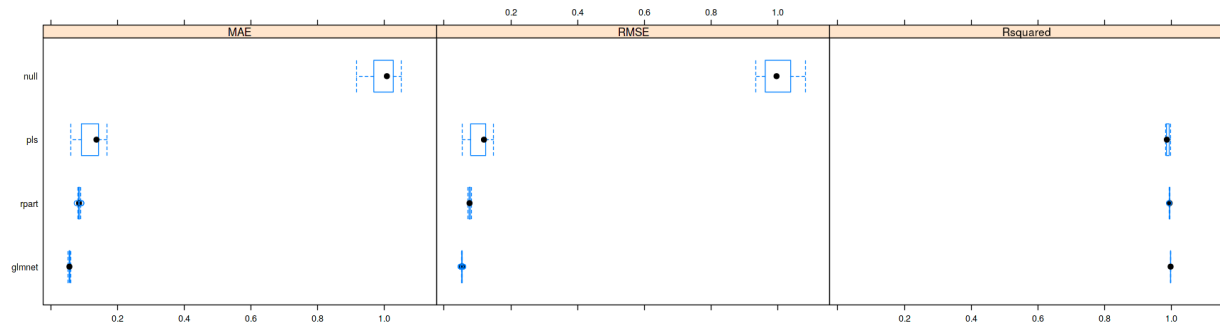


Figure 1: The models that were successfully trained.

The set of good candidate models were the top 3 performing models. These were fine tuned with the following preprocessing variations.

- Centring and scaling,
- Dimensional reduction: ica,pls,pca
- Bag imputation

From this set of good models, the best model was determined.

Best model

Based upon RMSE, the best model is **rf**, however it is not significantly better than **RRF** and **Rborist**. Any of these would perform well.

Performance on unseen data

The test data was predicted using the best model. It generated the chart as seen in Figure 2.

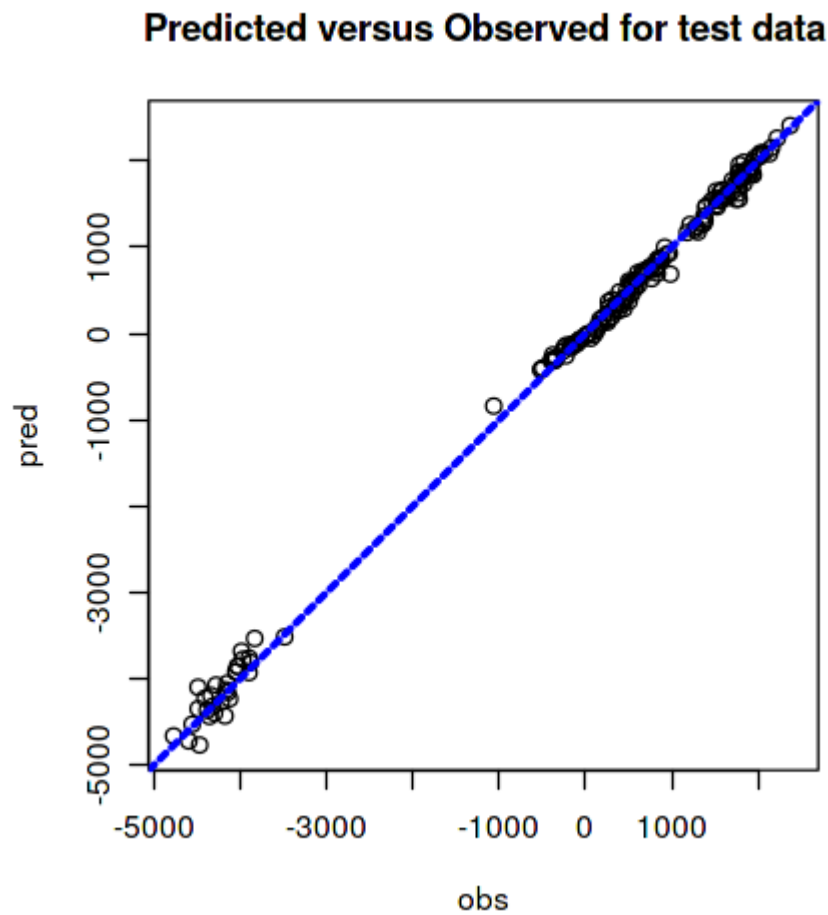


Figure 2: Test data predictions using the best model.

The expected performance on unseen data for the best model, **rpart**, can be seen in Table 2:

Table 2: Expected performance on unseen data for the best model, rpart.

Test_Metric	Value
RMSE	12.9000
MAE	21.9000
R2	0.9996

Observation that do not fit the model

The model-based outliers (at IQR Multiplier of 2.3) are shown in Figure 3:



Figure 3: The model-based outliers (at IQR Multiplier of 2.3).

Method description

The best model uses method **rpart**.

There are sufficient observations for this method to be a good fit. The method works by <your explanation here> <your references here>

This method seems to work well with this data because the data has a non-linear relationship with the predictors. Since **rpart** can handle nominal variables (of low cardinality) natively, no dummy encoding was necessary. However, both with and without dummy encoding were explored. While **rpart** can tolerate missing values, the missing values were imputed ahead of this method.

Transparency

The best model, **rf**, is not particularly transparent. It would not meet the criterion for being able to explain how the variables affect the outcome of the model. Fundamentally this is because random forest models are ensembles of many tree models, which then obscures the simplicity of the tree models.

In order to achieve transparency we would need to revert back to model the **glm**.

There are sufficient observations for this method to be a good fit. The method works by <your explanation here> <your references here>

Since **rf** out performed **glm**, we know that some non-linearities must be holding **glm** back. The **glm** model was subjected to some pre-processing fine tuning. In order to introduce some non-linearity, interactions of the predictors were introduced in the pre-processing. This revealed an important interaction between “District” and “Month” that improved the **glm** model to be (statistically) similar to the **rf** model.

The expected performance on unseen data for the best transparent model, **glm**, can be seen in Table 3:

Table 3: Expected performance on unseen data for the best transparent model, glm.

Test_Metric	Value
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RMSE	16.90
MAE	24.80
R2	0.98

The End